

GetSnap User Manual

Connect an instrument and capture screen images / waveform data with a single hotkey or button press, then save, copy, or paste them.

[Instrument] --Cable(LAN/USB/GPIB)--> [GetSnap] --Hotkey/Button--> [Image/Data] --> [Save to Folder / Clipboard / Excel / CSV]

The screen-capture commands are initial defaults based on each vendor's commonly published command set. They may differ by model/firmware. You can adjust them directly from the app - see "8. Adding & Editing Instruments" below.

1. Launching the App

Method	Action
exe file (recommended)	Double-click dist \ GetSnap.exe
From Python	venv \ Scripts \ activate → python app.py

GPIB connections require NI-VISA or Keysight IO Libraries Suite (see "9. Troubleshooting").

2. Screen Layout

Tab	Contents
Capture	Capture buttons (save to folder / copy to clipboard / paste into Excel / save CSV data), Legend, DIV Conditions, preview, history
Devices	VISA resource list, connect
Settings	Save location, background mode, hotkeys, Excel paste, instruments, data export (CSV), Legend, DIV Conditions, language

 Save to Folder

 Copy to Clipboard

 Paste into Excel

 Save Data (CSV, etc.)

☒ Enable Legend (applies to folder save & clipboard copy)

MODEL	
Ope. Mode	
BOM	
Condition	
Note	

☒ Enable measurement conditions (shows enabled channels' voltage range on the right of the image)

Legend to Display in the Image

CH1	Vds	V		
CH2	Vgs	V		

+ Add Channel



No Snap

History

Detected VISA Resources

↻ Refresh

✓ Connect

Manually enter a resource string (if not auto-detected)

e.g. GPIB0::7::INSTR

→ Manual Connect

Connected: TESTSCOPE,MODEL,SN001,FW1.0 [TestScope]

3. Basic Operation

Step	Action
① Connect	"Devices" tab → "Refresh" → select target → "Connect"
② Capture	Hotkey, or the buttons on the "Capture" tab

③ Check	Review the preview/history on the "Capture" tab
④ Use it	Save to folder / copy to clipboard / paste into Excel / save CSV data (see "5. Pasting into Excel" and "6. Saving CSV Data")

Hotkeys

The "Settings" tab lets you turn each of the three hotkeys on or off individually, and freely reassign the key combination for each (defaults below). If a hotkey conflicts with another app, just turn that one off and use the buttons on the "Capture" tab instead.

Purpose	Default key
Save to folder	Ctrl+Alt+F1
Copy to clipboard	Ctrl+W
Paste into Excel	Ctrl+Alt+F2

When an Error Occurs

The [①...] shown at the start of the error dialog indicates which step failed.

Step shown	Meaning
Instrument not connected / Preset not registered	Not yet connected on the "Devices" tab, or no preset is registered
①Connecting to VISA resource / ①Sending command / receiving response	Communication issue (cable, address, VISA runtime)
②Identifying model / ②Decoding image	Response data format issue
③Post-processing	Image processing issue (background conversion, Legend/DIV Conditions compositing)
④Saving / ④Copying	Permissions or disk space issue at the save destination
Data export not supported	"Save CSV Data" was pressed, but the connected model has no data export commands registered

① Sending data-export command to instrument / receiving response	Communication issue during CSV data export (cable, commands)
② Saving data file	Data retrieval succeeded; issue with permissions or disk space at the save destination

4. Adding Information to the Image: Legend & DIV Conditions

Two toggles on the Capture tab let you auto-composite information onto the image. Both apply to folder save, clipboard copy, and pasting into Excel alike.

☒ Enable Legend (applies to folder save & clipboard copy)

MODEL	
Ope. Mode	
BOM	
Condition	
Note	

☒ Enable measurement conditions (shows enabled channels' voltage range on the right of the image)

Legend to Display in the Image

CH1	Vds	V		
CH2	Vgs	V		

+ Add Channel

4.1 Legend: free-text fields attached to the image

Turning on "Enable Legend" shows a list of items on the Capture tab (e.g. MODEL, Ope. Mode, BOM, Condition, Note). Fill in values and capture - they're composited as a table at the top of the image. Adding/removing items, reordering, and font settings are managed from the "Legend" section of the "Settings" tab (not covered by "8. Adding & Editing Instruments").

4.2 DIV Conditions: per-channel voltage range, shown automatically

Turning on "Enable measurement conditions" lets you pick which channels (CH1-8) / Math traces (M1-8) to display, and enter a custom legend name per row (e.g. CH1: Vds). On every capture, GetSnap queries the instrument and composites the actual voltage range (e.g. 5 V/div) into a panel whose row backgrounds match each channel's waveform color.

Action	Steps
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Add a channel	" + Add Channel" → choose CH1-8 / M1-8 from the dropdown
Edit the legend name	Type into each row's text field (leave blank to show just the ID, e.g. "CH1"). Press "Enter" or "Shift+Enter" to add line breaks for a multi-line legend name
Edit the unit	Type directly into each row's unit field (default V). For a current probe, for example, enter A - any unit other than voltage can be entered freely (there is no way to query the unit from the instrument, so it's always set manually)
Remove a channel	Red trash-can icon at the right of each row
Change the insertion position	"Settings" tab → "DIV Conditions Position" (Right / Left / Bottom / Top)
Change the font	"Settings" tab → "DIV Conditions Font"

Choosing "Top" inserts the DIV Conditions panel between the Legend and the waveform image (stacking order: Legend → DIV Conditions → waveform).

Example output (CH1=Vds, CH2=Vgs shown on the right):



4.3 Choosing channel colors (always manual)

Click the color button (the square swatch to the right of each legend-name field) to pick a color from the palette manually (hex input also supported). Any channel you haven't given a color falls back to the default palette (CH1=yellow, CH2=green, etc., matching common oscilloscope conventions).

Automatically reading display colors from the instrument was removed: it required more commands when adding a new supported instrument and didn't generalize well across vendors. Colors are always set one of the two ways above.

4.4 If the voltage range query fails

If reading a channel's DIV condition voltage range fails, the capture itself is never aborted. An error message is shown, and the image is saved/copied without the DIV Conditions panel (the same appearance as when the feature is disabled). If you keep seeing this error, check the command setup in the "Support DIV condition display" card - see "8.7 DIV Conditions commands".

4.5 Adding the time axis (horizontal) range

Turning on "Enable time axis display" under "Settings" tab → "DIV Conditions" adds a row for the time axis (horizontal) range (e.g. 1 ms/div) at the front of the DIV Conditions panel.

Action	Steps
Turn time axis display on/off	"Settings" tab → "DIV Conditions" → "Enable time axis display"

Non-oscilloscope instruments (VNAs, spectrum analyzers, etc.) don't have a time-based horizontal axis, so while "Enable time axis display" is OFF, GetSnap never sends a time axis range command to the instrument (to avoid errors on instruments that don't support it). The command is only sent once you turn it ON, and if the query fails, the capture itself is never aborted (same as 4.4 - an error is shown and the image is saved without the time axis information). You can check the command setup in "8.7 DIV Conditions commands".

5. Pasting into Excel

Pressing "Paste into Excel" (or its hotkey) inserts the image as a picture shape directly at the currently selected cell in Excel. Unlike a clipboard-based paste, the shape size is specified in points, so it's unaffected by the sheet's zoom level, and it's always generated from the full-resolution image right after capture, so there's no quality loss.

Setting	Details
Image width when pasted into Excel	"Settings" tab → "Excel Paste" section (default 80mm; height is computed automatically to preserve the original aspect ratio)

If Excel isn't running, or no cell is currently selected, nothing happens (when triggered by hotkey this is only a brief status-bar note - no error dialog is shown).

6. Saving CSV Data

Pressing the green "Save Data (CSV, etc.)" button on the "Capture" tab runs the connected model's waveform data commands and saves the result as a separate CSV file, alongside the image, in the same save folder.



6.1 Automatically fetching every enabled channel

If the connected model's preset has a "channel enabled-check command" configured, GetSnap automatically detects every channel currently displayed (enabled) on the oscilloscope and combines them into a single CSV (SampleIndex, CH1, CH2, ... columns). Models without this command configured fetch CH1 only (see "8.6 CSV data export commands").

6.2 Automatic decimation for long record lengths

On models with long record lengths, the CSV file size can reach tens of MB or millions of rows - too large to open in Excel (max ~1.04M rows per sheet). Only when the point count exceeds the "Max Points for CSV Export" setting (default 50,000; "Settings" tab → "Data Export") does GetSnap automatically decimate it, while preserving each channel's and each segment's minimum and maximum values. Whether or not decimation happens, a `SampleIndex` column (the original sample position) is always included, so the time-series information is never lost.

6.3 Cancelling a save in progress

While saving, the status bar shows an animated progress indicator and a "X" button. On models with long record lengths, retrieval can take tens of seconds to a few minutes - press "X" if you want to cancel.

VISA communication has no safe cancellation mechanism, so pressing "X" forcibly disconnects the instrument to interrupt the operation. After cancelling, reconnect from the "Devices" tab.

7. Supported Instruments

The actual commands can always be reviewed and edited from the dialog in "8. Adding & Editing Instruments" below.

Vendor	Image Format	CSV Data Export	Notes
Tektronix	PNG	Yes	-
Yokogawa	PNG	Yes	Return-to-local uses GTL only (verified on real hardware, see 8.5)

To keep debugging focused, the built-in presets are currently limited to these two vendors. Other instruments can be added via "8.3 Adding a new instrument".

8. Adding & Editing Instruments

Go to "Settings" → "Instrument Settings..." to add or edit supported instruments. Your edits persist across app restarts.

Save

Save Folder

C:\Users\daisu\AppData\Local\Temp\getsnap_manual_screens_ja

Browse...

Background Conversion Mode

Invert (B/W)

Hotkeys

Same actions are also available from the buttons in the "Capture" tab

☒ Save to Folder

Ctrl+Alt+F1

☒ Copy to Clipboard

Ctrl+W

☒ Paste into Excel

Ctrl+Alt+F2

Excel Paste

Image Width When Pasted into Excel

80.00 mm

Instruments

See Manual_ENG.pdf for the list of supported vendors and commands.

Instrument Settings...

Data Export

On models with long record lengths, the file size may become too large to open in Excel.

Max Points for CSV Export

50000

Legend

Legend items (edit label / remove)

MODEL

Ope. Mode

BOM

Condition

Note

+ Add Item

Legend Font

メイリオ (Meiryō)

15

DIV Conditions

DIV Conditions Position

Right

DIV Conditions Font

メイリオ (Meiryō)

15

☐ Enable time axis display

Display

Language / 言語

日本語

8.1 First, find the model match string

Before adding a new instrument, connect to it once from the "Devices" tab. You can connect even without a registered preset - the `*IDN?` response string is shown as-is on screen (e.g. `Connected: ANRITSU,MS2840A,SN001,FW2.0 [Unsupported model (no preset registered)]`). Use part of that string (e.g. the vendor name) as the "model match string".

8.2 Editing an existing preset

Select an instrument from the list to open its form on the right. Edit the fields and press "Save" to override that instrument's settings only. Built-in presets cannot be renamed. "Reset to Default" restores the original content at any time.

Registered instruments

Tektronix

Yokogawa

+ Add New

This is a built-in preset. Its name cannot be changed (use "+ Add New" to register it and overrides this instrument's settings only).

[Required] Name

Yokogawa

[Required] Model match string (substring of the *IDN? response, matched case-insensitive)

yokogawa

[Required] Image save commands (one per line, last line is the query)

:IMAGe:FORMat PNG
:IMAGe:SEND?

[Required] Commands (SCPI, one per line)

Sent in addition to the VISA-level GTL command when returning the instrument to local connections already use GTL as the primary mechanism, so this can usually be left blank.

Optional

✓ Support no-invert capture

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave instead.

[Optional] Commands (one per line, last line is the query)

:IMAGe:FORMat PNG
:IMAGe:SEND?

Image Format

Delay Between Con

PNG

0.00 s

✓ Support CSV data export

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled displayed (enabled) on the oscilloscope is automatically fetched and combined into a :

[Optional] Data commands (one per line, last line is the query; "{ch}" is replaced with t

:STOP
:WAVeform:TRACe {ch}

[Optional] Data export format (e.g. CSV)

CSV

[Optional] Channel enabled-check command ("{ch}" is replaced with the channel num

:CHANnel{ch}:DISPlay?

8.3 Adding a new instrument

Press "+ Add New" at the bottom left to open a blank form. Fill in the required fields (name, model match string, image save commands, return-to-local commands) and press "Save" to add it to the list. The red "Delete" button removes it at any time.

Optional fields are grouped into three feature cards - "Support no-invert capture", "Support CSV data export", and "Support DIV condition display" - and each card's fields only appear once you turn on its checkbox for a feature you actually want to use. Leave a card off and its fields stay hidden, keeping the form focused on what you need.

Registered instruments

Tektronix

Yokogawa

+ Add New

[Required] Name

[Required] Model match string (substring of the *IDN? response, matched case-insensitive)

e.g. yokogawa

[Required] Image save commands (one per line, last line is the query)

e.g.
:IMAGe:FORMat PNG
:IMAGe:SEND?

[Required] Commands (SCPI, one per line)

Sent in addition to the VISA-level GTL command when returning the instrument to local control after a capture or data export. GPIB/USB connections already use GTL as the primary mechanism, so this can usually be left blank.

Optional

☐ **Support no-invert capture**

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave this off to reuse the image save commands instead.

Image Format

PNG

Delay Between Commands

0.00 s

☐ **Support CSV data export**

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled-check command, every channel currently displayed (enabled) on the oscilloscope is automatically fetched and combined into a single CSV.

☐ **Support DIV condition display**

Used by the "DIV Conditions" feature to auto-reflect the user-selected channels'/Math traces' voltage range (V/div) in the image.

[Optional] Notes

Save

Delete

8.4 Field Reference (required)

Field	Description	Example
Name	Display name in the list (built-in presets cannot be renamed)	Anritsu
Model match string	Substring of the *IDN? response, matched case-insensitively; the first match from the top of the list is used	anritsu
Image save commands	Screen capture commands. One per line; the last line is the query that returns the image	:HCOPY:SDUMP:DATA?
Return-to-local commands	Extra command(s) to restore the front panel after capture / data export. Usually left blank (see 8.5)	Usually blank

Image format and Delay between commands are always-valued inputs (combo box / spin box), so they carry no required/optional tag, but both are still used for image capture.

8.5 Leave "Return-to-local commands" blank in most cases

For GPIB/USB connections, GetSnap already restores the front panel automatically after capture via VISA-level GTL (Go To Local). Entering a command the instrument doesn't recognize here can actually cause a command-error message or leave the panel locked (confirmed through real hardware testing on a Yokogawa DLM5058). Only enter a command here - one you've verified against the model's programming manual - if GTL alone does not restore the panel for your instrument.

8.6 CSV data export commands (optional)

Turning on "Support CSV data export" reveals the following fields.

Registered instruments

Tektronix

Yokogawa

+ Add New

yokogawa

[Required] Image save commands (one per line, last line is the query)

:IMAGe:FORMat PNG
:IMAGe:SEND?

[Required] Commands (SCPI, one per line)

Sent in addition to the VISA-level GTL command when returning the instrument to local connections already use GTL as the primary mechanism, so this can usually be left blank.

Optional

✓ Support no-invert capture

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave instead.

[Optional] Commands (one per line, last line is the query)

:IMAGe:FORMat PNG
:IMAGe:SEND?

Image Format

Delay Between Con

PNG

0.00 s

✓ Support CSV data export

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled displayed (enabled) on the oscilloscope is automatically fetched and combined into a :

[Optional] Data commands (one per line, last line is the query; "{ch}" is replaced with t

:STOP
:WAVeform:TRACe {ch}

[Optional] Data export format (e.g. CSV)

CSV

[Optional] Channel enabled-check command ("{ch}" is replaced with the channel num

:CHANnel{ch}:DISPlay?

Max channels to check (CH1 through this number)

4

✓ Support DIV condition display

Used by the "DIV Conditions" feature to auto-reflect the user-selected channels'/Math

Field	Description	Example
Data commands	Waveform data retrieval commands. One per line, last line is the query. {ch} is replaced with the channel number (1, 2, 3, ...)	:WAVeform:TRACe {ch} → :WAVeform:SEND?

numeric value of the SENS field (the per-division sensitivity), so you can still configure it with the same {ch} substitution as a regular channel.

9. Troubleshooting

Symptom	Fix
Instrument doesn't appear in the list over GPIB	Install NI-VISA or Keysight IO Libraries Suite
Connects fine but capture fails	Check the failing step at the start of the error, then review commands in "8."
Want to change the image background color	Toggle "Background Conversion Mode" on the "Settings" tab
A hotkey doesn't respond	May conflict with another app; turn just that hotkey off in "Settings" and use the buttons on the "Capture" tab instead
Front panel unresponsive after capture	See "8.5 Leave 'Return-to-local commands' blank in most cases"
DIV Conditions shows an error / doesn't appear	Check the command setup in "8.7 DIV Conditions commands". On error, the image is still saved without DIV Conditions
Time axis (horizontal) range shows an error / doesn't appear	Check that "Enable time axis display" is ON in the "Settings" tab, and that the time axis range query in "8.7 DIV Conditions commands" is correct (see 4.5). On error, the image is still saved without the time axis information
Channel colors don't match the real instrument	See "4.3 Choosing channel colors" - pick the color manually from each row's color button
CSV data export only fetches some channels	Check the channel enabled-check command and max channel count in "8.6 CSV data export commands"
Cancelling a CSV save left the instrument disconnected	This is expected (see 6.3) - reconnect from the "Devices" tab